



The Advantages of Neutron Tomography for Non-Destructive Imaging of Lithium-Ion Batteries

The Nukes

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Abstract

This literature review focuses on neutron imaging techniques used to visualize the physical structure of lithium ion batteries in operando. Lithium ion batteries are ubiquitous, used to power many portable devices around the world such as phones, laptops, and headphones. The batteries are also used in medical devices, drones, electric vehicles, and many other applications where high electrical power density and rechargeability are required. Because lithium ion batteries (LIBs) are so common, there is significant interest in improving their properties such as power density, safety, cost, and reliability. However, to research these advancements, the internal structure must be analyzed in a non destructive manner while the battery is in operation. The parts of an LIB are degraded through charge cycling of the battery, resulting in problems such as dendrite growth on the lithium anodes and gas generation due to decomposition of the electrolyte and cathode. Neutron radiography has been proven to be an effective method of monitoring these processes, as compared to conventional X-ray imaging techniques, neutrons are non destructive and capable of achieving much higher contrast imaging of the battery, as unlike X-rays, neutrons are easily attenuated by light elements such as lithium. While neutron imaging does not have as high resolution as X-rays, it significantly increases the amount of information that can be gathered from in operando studies of LIBs.

Motivations for Developing Neutron Imaging

The development of neutron imaging has been motivated by the need to address challenges in the industrialization of lithium metal electrodes with conventional liquid electrolytes, specifically when studying microscopic dendrite growth. Dendrite growth, in the form of mosses, needles, and globules, is attributed to the leading cause of short circuits and catastrophic failures in lithium batteries. [8] [7] [11]

Gas generation during battery lifespan, caused by electrolyte decomposition and structural release, poses performance and safety risks. Traditional imaging technologies face limitations in observing gas evolution, making nondestructive probes like neutron imaging with high-energy X-rays and neutrons essential for providing position information on degradation. [6] [7] [8] [10]

Usage of neutron imaging during battery discharge behavior reveals irreversible capacity reduction, structural deterioration, and safety hazards in lithium-ion batteries. NI techniques prove valuable in assessing lithium ion concentration, providing information for safe battery design and estimating battery life. Alongside these assessments, NI shows the importance in reducing the time consumed and the production costs of lithium ion batteries. These studies open perspectives for battery characterization under real operating conditions, especially for novel materials and designs. [6] [10] [11] [15]

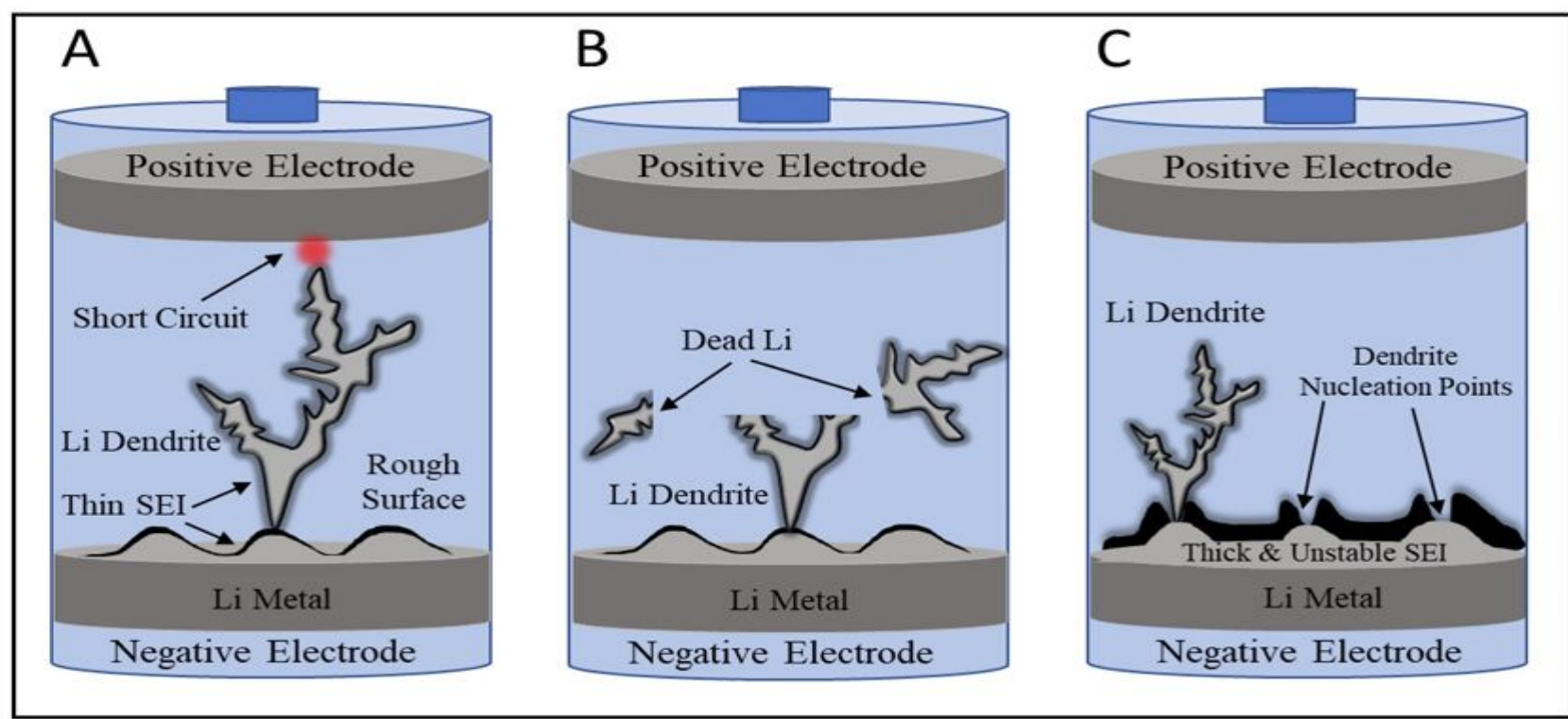
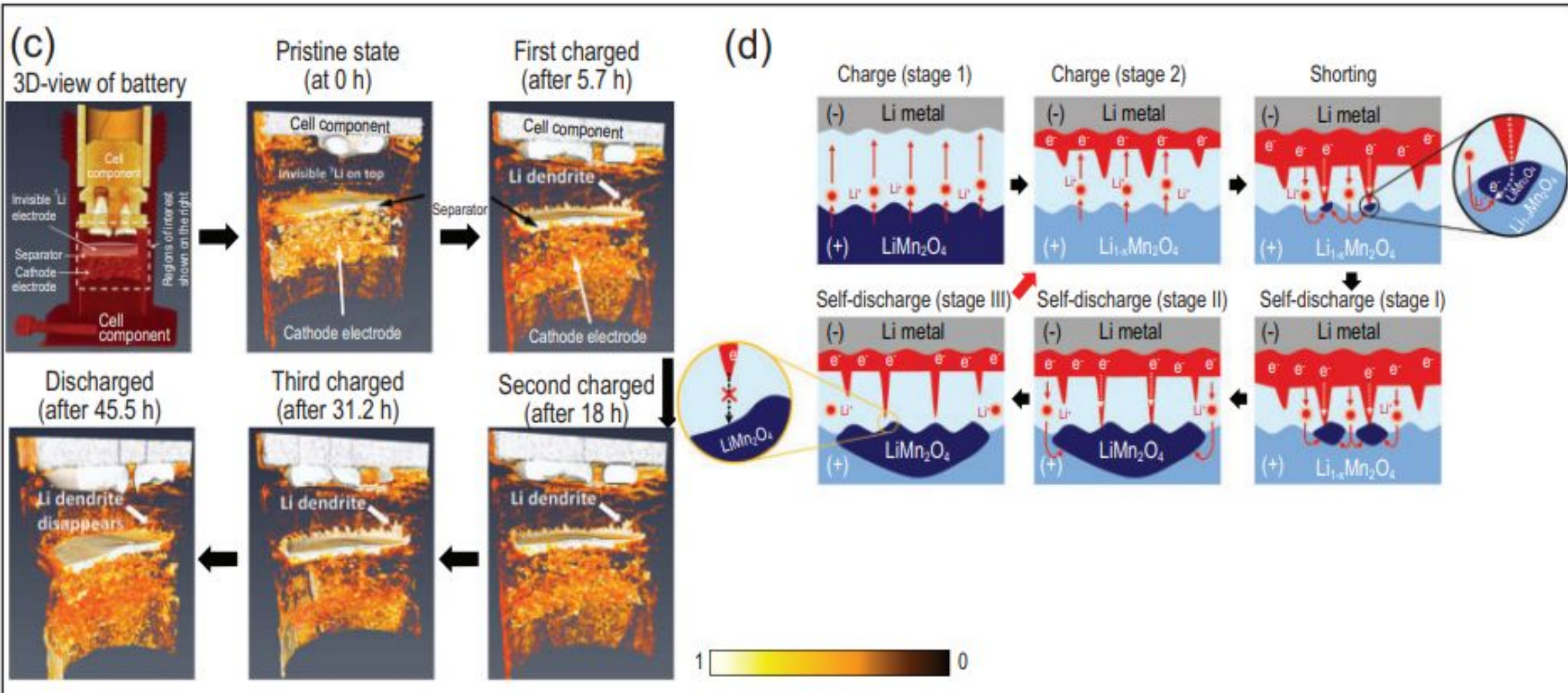


Fig. 1. Representative schematic of the main challenges associated with Li metal anode: (a) short circuit induced by Li dendrites, (b) formation of inactive (dead) Li, and (c) development of thick and mechanically unstable SEI.

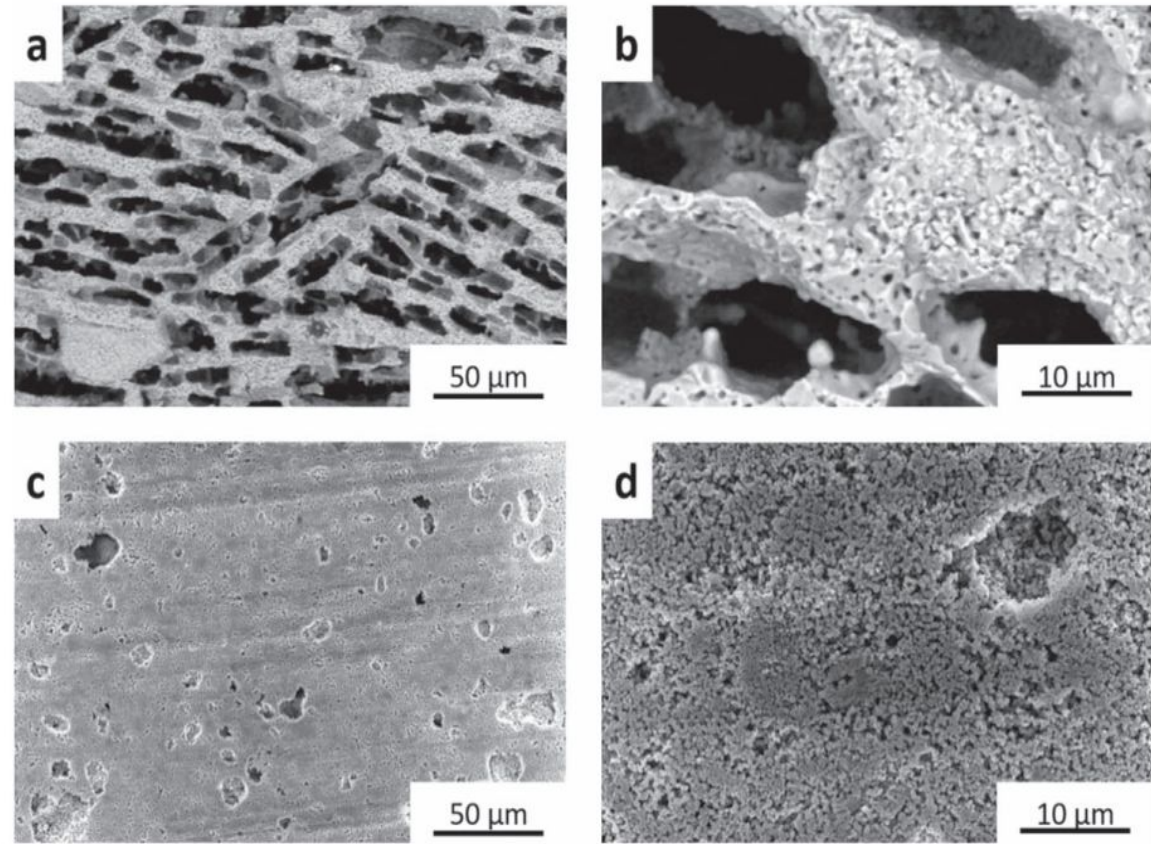


Operando neutron depth profiling (NDP) has been employed to investigate dendrite formation mechanisms in solid electrolytes. Real-time visualization using NDP reveals the direct deposit of lithium dendrites inside specific solid electrolytes, emphasizing the importance of electronic conductivity in dendrite formation. [7] [19]

Despite decades of theoretical modeling, experimental evidence suggests the complexity of Li dendrite formation mechanisms extends beyond existing models. Advanced materials characterization methods, particularly in situ and operando techniques like NI, are essential for comprehensively understanding Li dendrite phenomena and overcoming challenges posed by Li's reactivity and sensitivity. [1] [5] [15]

Restrictions/Limitations of Neutron Imaging

Due to increased difficulty of detecting neutrons vs more conventional methods, getting the resolution required for some analysis can be lengthy or costly. [11] [21]



Due to the nature of neutrons, generating them in a laboratory setting is more expensive than other methods such x-rays [5]

“Although, NPD is [a] robust technique to depth profiling of Li concentration, it does not allow for direct visualization of Li metal microstructures.” [5]

Advantages of Neutron Imaging

Neutron imaging is crucial for gaining insights into lithium battery composition by discerning between Li isotopes (⁶Li and ⁷Li). The unique ability of neutrons to distinguish isotopes provides a better understanding of the source and structure of faults within the electrode. Neutron imaging, as a non-invasive technique, offers high penetration depth, making it the best option for studying lithium batteries during charging and discharging. [1] [2] [4] [5] [6] [11] [16]

In comparison to X-ray techniques, neutron imaging allows for non-destructive characterization of battery interiors. Neutrons interact with atomic nuclei, not electrons, making them highly penetrating and ideal for studying large objects like batteries. The visibility of neutrons for light elements such as hydrogen and lithium enables direct observation of phenomena like lithium diffusion, electrolyte consumption, and gas formation. [1] [3] [6] [11]

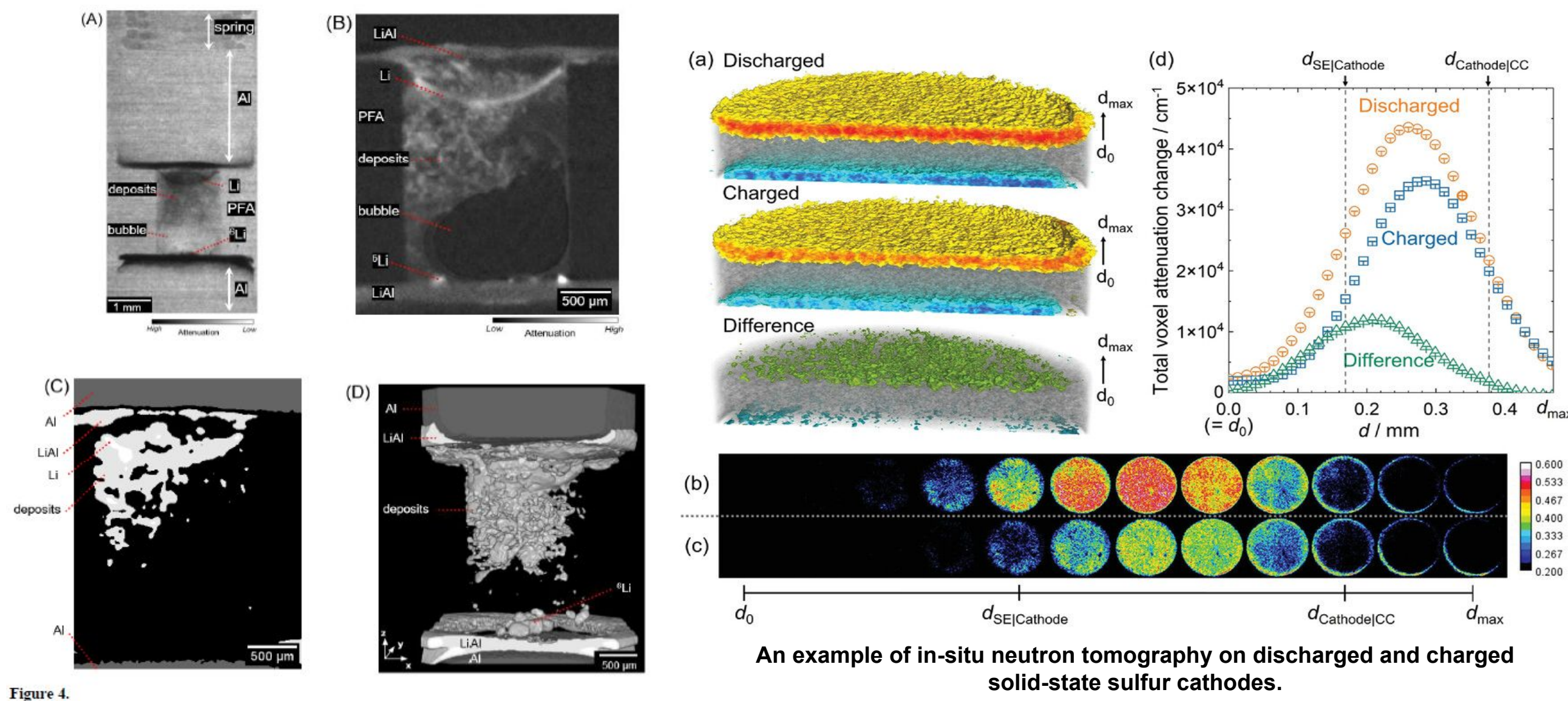


Figure 4. 2D neutron radiography (A) and 3D tomography (B) of ⁶Li-Li Cell and refined 2D (C) and 3D (D) images of tomography.

The isotopic sensitivity of neutron techniques allows for enormous contrast adaptability and dynamic investigation of materials evolution, particularly in Li batteries. Neutron imaging is particularly suitable for visualizing the 1-5 micron lithium dendrites, which due to their small size have been difficult to analyze in the past. The technique's high visibility for hydrogen and functional makes it effective in tracking lithium exchange processes within batteries. [3] [6] [11] [22]

Non-destructive neutron-based methods, such as neutron radiographic imaging and neutron depth profiling, offer valuable insights into electrochemical reactions, Li dendrite formation, interfacial transport behavior, and the kinetics of Li transport in battery cells. The high sensitivity of neutron imaging, compared to X-ray imaging, allows for the investigation of Li plating behavior inside solid electrolytes and the visualization of Li concentration profiling in battery cross-sections. [5] [12] [22]

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Comparison to Similar Imaging Methods

Neutron tomography is effective for in situ analysis and capturing the morphology of Lithium deposits. Experiments have been completed that demonstrate the ability to measure deposit volumes comparable to X-ray techniques. For higher resolution, contrast, and shorter acquisition times there will need to be further optimizations. Currently there are proposed improvements being made that include neutron-transparent buffering of the metal layer and the ability to image a reduced cell diameter. [11]

Neutron imaging and X-ray tomography reveal coral-like pore structures in lithiated graphite electrodes through common orthogonal slices of their respective tomography. Neutron imaging enhances contrast but has been found to not always lead to higher resolution when images are reconstructed. [16] [22]

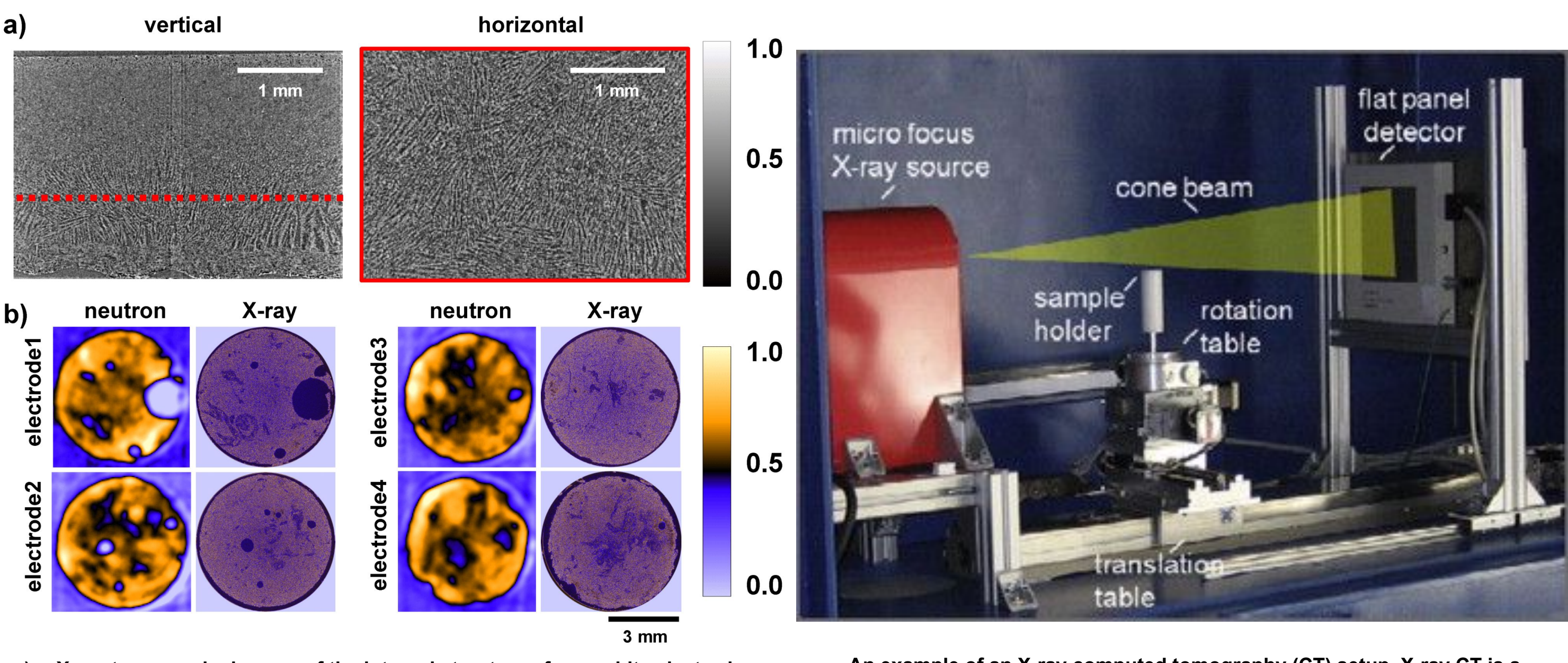


Figure 4. (a) X-ray tomography images of the internal structure of a graphite electrode. (b) Comparison between neutron and X-ray imaging of a lithiated electrode. (c) An example of an X-ray computed tomography (CT) setup. X-ray CT is a common X-ray imaging technique.

Neutron imaging has advanced complementary to X-ray techniques, especially in battery materials characterization. In X-ray synchrotron applications beam damage can be problematic whereas neutron imaging does not cause any measurable radiation damage. Neutron-induced sample activation may pose logistical challenges but is usually short-lived. [21]

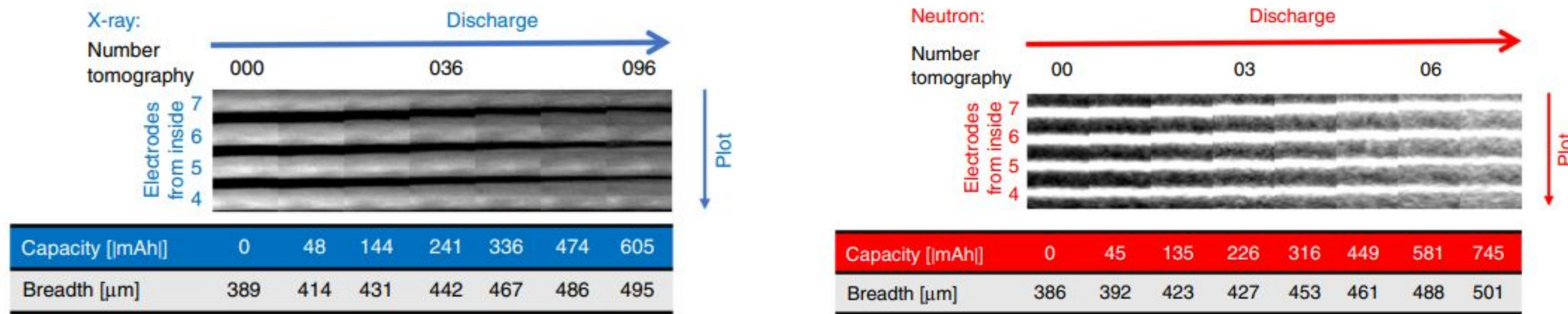
Neutron radiograms and X-ray tomography both highlight higher neutron absorption in polymeric foams. Enhanced contrast in neutron imaging does not necessarily guarantee higher resolution in reconstructed tomography. [16]

In-situ TEM is a major breakthrough in battery research and had proven reliable for studying Lithium dendrites due to high spatial resolution and lower beam damage. Utilizing for dendrites does have disadvantages which can lead to deviations from commercial cell conditions and potentially invalid conclusions. [5]

Synthesis with Other Methods

Currently electron and X-ray techniques rely on the atomic number of the atom for contrast but have difficulty with heavier transition metal elements. Neutron scattering addresses these limitations and is helping to fill the gaps in accurately describing the structure's response to Lithium batteries. [2]

Combining X-ray and neutron imaging provides valuable benefits for interpreting materials with different attenuation properties. This provides a versatile approach and is beneficial for optimizing various battery chemistries and applicable rechargeable cells. [20]



Nondestructive neutron imaging compensates for X-ray deficiencies in analyzing Lithium metal. This imaging allows the observation of Lithium electrodeposits which enables monitoring of the metal dendrite growth and volume changes. [6]

Custom built Li-ion batteries offer a thorough analysis using diverse techniques as well as a higher resolution in neutron powder diffraction when synchronized with X-ray diffraction radiography. [15]